

GIK INSTITUTE OF ENGINEERING SCIENCES & TECHNOLOGY
FACULTY OF ENGINEERING SCIENCES

Syllabus/Course Outline of ES 341 (Numerical Analysis)

Spring Semester 2025

Course Code : ES 341
Course Title : Numerical Analysis
Credit Hours : 3
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Course Introduction				
This course is required for engineering students. The pre-requisite is the Differential Equations and linear Algebra taught in third semester. The aim of this subject to develop the understanding of numerical techniques for solving linear and nonlinear equations and differential equations of real world engineering.				
Course Contents				
• Mathematical Preliminaries and Errors Analysis • Iterative methods of root finding • Solution of system of linear equations • Interpolation and polynomial approximation • Numerical Differentiation and Integration • Numerical Solution of Ordinary Differential Equation				
Mapping of CLOs and PLOs				
S.No	CLOs	PLOs	Bloom Taxonomy	
After the end of this course, students will be able to				
CLO1	Apply different polynomial approximation methods	PLO1	C3	
CLO2	find the roots of linear and nonlinear equations	PLO1	C3	
CLO3	Use various numerical methods for differentiation, integration and solution of ordinary differential equations	PLO1	C3	
CLO4	Write efficient, well-documented code and present numerical results in an informative way.	PLO1	C3	
Direct Assessment Tools Based on CLOs				
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	30%	20%	20%	10%
Assignments	20%	20%	20%	50%
Midterm Exam	30%	30%	30%	20%
Final Exam	20%	30%	30%	20%
Grading Policy				

Assessment Items	Weightage
Assignments	10%
Announced Quizzes	20%
Midterm Exam	30%
Final Exam	40%
Text and Reference Books	
Text Book: Numerical Analysis (9 th edition) by R. L. Burden and J. D. Fairs, Books/Cole	
Reference Books: 1. Numerical Analysis by D. Kincaid and W. Cheney 2. Numerical Methods, for Computer Science, Engineering and Mathematics by John H. Mathew	
Administrative Instruction	
• Student Attendance is expected to be 100%, and minimum 80% (mandatory) attendance that is required to sit in the final exams • Student must pay the attention for reading the text books chapter for course assessment • All the direct assessment tools i.e., Quizzes, Assignment, Midterms, Project and final Exams must be attempted. Failure to attempt in any of the assessment tools without any medical reasons may results to fail in that particular assessment. All quizzes would be taken together for all sections. • Students are advised to study the previous lecture before next class for better understanding. • Class participation is highly encouraged. It develops more interest. Students are also advised to spare some time for group discussion with their classmates to explore new ideas. • For any query please contact instructor during office time. • All quizzes/examinations will be closed book. No graphical calculators will be allowed. • There will be five announced quizzes. Out of these five, four best quiz results of each student will be counted for grading purpose. • Assignments must be submitted as per instructions mentioned in the assignments. • In any case, there will be no retake of scheduled/surprise quizzes.	
Lecture Breakdown	
Lecture 01	Introduction to Numerical Analysis, Review of Calculus
Lecture 02	Types of errors. Measuring size of errors, Round-off errors and floating point arithmetic. Chopping and rounding
Lecture 03	The Bisection Method and related theorems, Numerical Examples using Bisection Method
Lecture 04	Fixed point iterations and related theorems, Numerical Examples using Fixed point iterations
Lecture 05	The Newton and Secant Methods and their applications
Lecture 06	Error Analysis and Modified Newton's Method with examples
Lecture 07	Continue Lecture 06
Lecture 08	Multiple roots, Horner's Method and Algorithms for Iterative Techniques
Lecture 09	Introduction to matrices and Linear System of Equations

Lecture 10	Gaussian Elimination with Backward Substitution and Gauss-Jordan Method, Pivoting Strategies, Partial and Scaled Partial Pivoting
Lecture 11	LU Factorization
Lecture 12	Interpolation and Lagrange Polynomial, Taylor Polynomial
Lecture 13	Newton's Forward and Backward divided differences and Polynomials
Lecture 14	Newton's Forward and Backward differences
Lecture 15	Centered divided difference and Polynomials
Lecture 16	Hermite interpolation
Lecture 17	Splines, Construction of Splines (Natural and Clamped Splines)
Lecture 18	Applications of Splines, Algorithm for interpolation
Lecture 19	First order derivatives using two, three and five points
Lecture 20	Formulas for higher derivatives and Round-Off Instability
Lecture 21	Richardson's Extrapolation
Lecture 22	Trapezoidal & Composite Trapezoidal rules rules of integration and examples
Lecture 23	Simpson's Rule of integration and examples
Lecture 24	Composite Simpson's Rule of integration and examples
Lecture 25	Romberg integration
Lecture 26	Algorithm for Numerical Differentiation and Integration
Lecture 27	Review of Ordinary Differential Equations, Initial and Boundary Conditions, Euler's Method for the Solution of well-posed initial-value Problem
Lecture 28	Taylor's Method of Orders two and Four
Lecture 29	Modified Euler and Runge-Kutta Method of Order Four Methods
Lecture 30	Algorithm for Numerical Solution of Ordinary Differential Equations

Note: This outline serves only as a rough guidance of the course. It may be changed or modified as and when deemed necessary by the instructor.